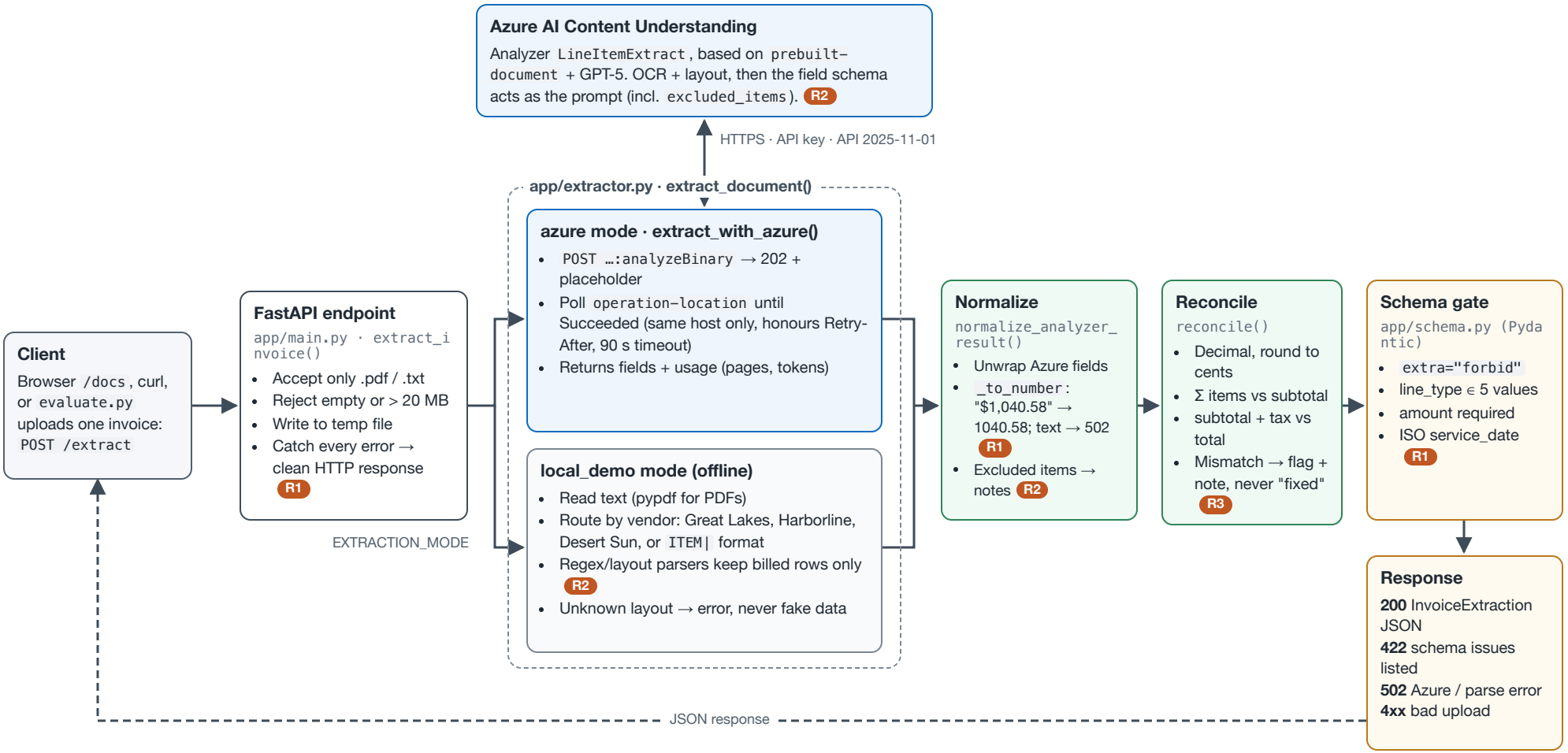
Fleet Invoice Analyzer: Architecture

FastAPI service that turns a repair-shop invoice (PDF/TXT) into schema-valid JSON of **billed** line items, with totals reconciled in code.



Offline tooling: set-up, evaluation, tests

Analyzer set-up

`scripts/create_analyzer.py`

Reads `azure/analyzer_definition.json` (field schema: header, `line_items`, `excluded_items`, totals) and PUTs it to Azure (`--replace` overwrites), then polls until ready.

Evaluation R4

`scripts/evaluate.py --mode azure|local_demo`

Sends all 12 sample files through the real `/extract`, scores against `samples/labels.json`: line-item P/R/F1, headers, totals, reconciliation flag, leaked unbilled items → `evaluation/results_*.json`.

Test documents R4

`samples/ · scripts/make_sample_pdfs.py`

3 provided invoices (inv_07–09) + 3 created ones (inv_10 Pinecrest: declined rows, inv_11 Ironbridge: wrong subtotal, inv_12 Northstar: 2 pages, 2 taxes, deposit), each as .txt and .pdf.

Unit tests (pytest, 20)

`tests/`

Reconciliation maths, Azure polling (202 → Succeeded), number conversion and clean errors, excluded-item notes, every sample parses the same from .txt and .pdf, unsupported layout refused.

R1 schema-valid; bad output handled, no crash · R2 billed items only; recommended/declined excluded · R3 totals reconciled in code, mismatches flagged · R4 samples + 3 created docs evaluated